

Kishore Namburi

Data and AI Architect | Generative AI Systems

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PROFESSIONAL SUMMARY

Senior Engineering leader with 20+ years designing and delivering enterprise-scale cloud platforms for Fortune 500 financial institutions. Hands-on expertise in agentic AI systems and Model Context Protocol (MCP), combined with a strong track record of leading engineering teams, driving cloud migrations, and developing deterministic GenAI systems that eliminate operational bottlenecks. Combines elite software engineering practices with deep domain expertise in financial technology and merchant services to deliver production-grade AI solutions.

CORE COMPETENCIES

Agentic AI & GenAI Systems (MCP) • Cloud-Native Architecture • Enterprise Data Platform Design • Data Lakehouse & Mesh Patterns • Cloud Migration Strategy • Real-Time Streaming (Kafka, Spark) • Data Governance & Compliance • Engineering Team Leadership • Budget Ownership • Stakeholder Communication

TECHNICAL SKILLS

AI & Emerging Tech: Agentic and GenAI Systems (MCP), LLM Integration, Multi agent Orchestration, Guardrail Policy Design

Cloud Platforms: AWS (EMR, Glue, ECS, Lambda, S3, Athena, Aurora), Snowflake

Data Engineering: Apache Spark, Kafka, ETL/ELT Pipelines, Data Lakes, Real-Time Analytics, MPP

Programming: Python, PySpark, Java, SQL, Unix Shell

Data Management: Data Governance, Data Quality, MDM, Logical & Physical Data Modeling, Erwin

Leadership: Technical Strategy, Team Building (25+), Budget Management (\$2M+), Executive Communication

CERTIFICATIONS

AWS Certified Solutions Architect – Associate (SAA)

AWS Certified Data Analytics – Specialty

AWS Certified Database – Specialty

PROFESSIONAL EXPERIENCE

MLTek AI

Jul 2026 – Present

Founder & Principal AI Architect — Paytek (paytekai.online)

Remote

- Architected Paytek AI, a multi-agent platform automating merchant onboarding and risk profiling by converting unstructured conversations into structured, compliance-ready data.
- Designed a config-driven policy pipeline across 13 auditable policy types — threat detection, privilege elevation, and LLM-judge checks — enforcing deterministic execution for financial writes.

- Built a compliance pipeline with real-time PII redaction, role-based access control, and an append-only audit store supporting forensic and cross-session tracing.
- Cut redundant LLM calls on cleared/read-only paths and added token-phase tracking and tool-call deduplication, reducing orchestration latency.
- Prototyped retrieval strategies for grounding agent responses in domain knowledge — full-context injection vs. graph-scoped semantic retrieval — and adopted an ontology with hybrid graph/vector search after measuring the relevance/context-size trade-off.

JPMorgan Chase & Co.

Aug 2024 – Jun 2026

Vice President – Sr. Data Architect

Plano, TX

- Pioneered Agentic AI proof-of-concepts using Model Context Protocol (MCP), automating complex merchant workflows and reducing manual processing time by 40%.
- Architected cloud-native analytics platform processing 3M+ daily transactions using AWS (EMR, Glue, Snowflake, Aurora), reducing query response times by 60% and improving merchant services customer experience.
- Designed backend systems and orchestrated migration of 500K+ merchant customers with zero service disruption.
- Led WePay decommissioning initiative — gap analysis across all workflows and backend systems, informing modernization strategy that eliminates ~\$10M in annual technical debt.
- Architected AWS multi-region deployment (US-East / US-West) achieving 99.99% availability with global failover capability.
- Built end-to-end pipeline reconciliation framework achieving 99.95% data quality, adopted as a standard across 4+ business groups.

Amazon Web Services (AWS)

May 2022 – Jul 2024

Data Architect – Data Warehouse & MPP

Dallas, TX

- Served as trusted technical advisor to Fortune 500 enterprises for cloud migration engagements spanning data warehousing, MPP, and modern data and AI platforms.
- Delivered average 45% cost reduction for customers migrating from legacy systems to AWS, while improving analytics performance and ROI.
- Designed data integration frameworks enabling customers to process 100TB+ datasets with 300% performance improvements.
- Built event-driven, serverless, low-code/no-code proof-of-concept orchestration patterns reducing customer development cycles by 50%.
- Developed production-grade ETL/ELT pipelines in Python, PySpark, and SQL supporting real-time decision systems across financial services and healthcare.

JPMorgan Chase & Co.

Oct 2020 – Apr 2022

Big Data Engineering Lead

Plano, TX

- Owned technical architecture for finance product data platforms serving 50+ internal stakeholders and 100+ vendor integrations.
- Established enterprise data governance and modeling standards (Erwin) adopted across multiple lines of business; drove compliance-ready pipeline frameworks for risk reporting.
- Led agile engineering teams through backlog definition, sprint planning, and POC delivery for revenue-sharing use cases.
- Managed \$2M+ infrastructure budget, optimizing spend while maintaining performance SLAs.

Mindtree Ltd.

Jun 2011 – Sep 2020

Sr. Technical Manager

Warren, NJ

Key Clients: American International Group (AIG) • Association of American Medical Colleges

- Led 10TB+ on-premises data mart migration to AWS, reducing operational costs by 40%.
- Designed customer-360 platform using advanced data lineage and mapping, contributing to 20% revenue growth through improved customer insights.

- Built automated data validation tooling raising accuracy to 99.8% and reducing manual validation effort by 80%.
- Managed 25+ person cross-functional teams across concurrent engagements; maintained 95%+ client satisfaction and secured \$10M+ in contract renewals.

Maxim Technologies Inc.

Mar 2005 – May 2011

Technical Lead

Wilmington, DE

Key Clients: Citigroup Inc. • JPMorgan Chase & Co.

- Designed star schema data marts with MicroStrategy integration supporting 200+ business users with sub-second query response.
- Built data quality engine with automated profiling and alerting, ensuring 99.5% data integrity for live production systems.
- Achieved 75% improvement in batch cycle times through performance tuning: mentored 10+ developers, accelerating onboarding by 50%.

INDUSTRY EXPERIENCE

Banking & Financial Services • Healthcare & Life Sciences • Insurance & Reinsurance • Telecommunications • Merchant Services • Payment Processing

EDUCATION

Bachelor of Engineering — Andhra University College of Engineering, Visakhapatnam

ONLINE & PROFESSIONAL PORTFOLIO

- Professional Network: [linkedin.com/in/tonamburi](https://www.linkedin.com/in/tonamburi)
- Open Source & Code Repository: github.com/namburik
- Paytek Live AI App: paytekai.online